

SCIENTIFIC DISCOVERIES, NOBEL PRIZES AND SCIENTISTS

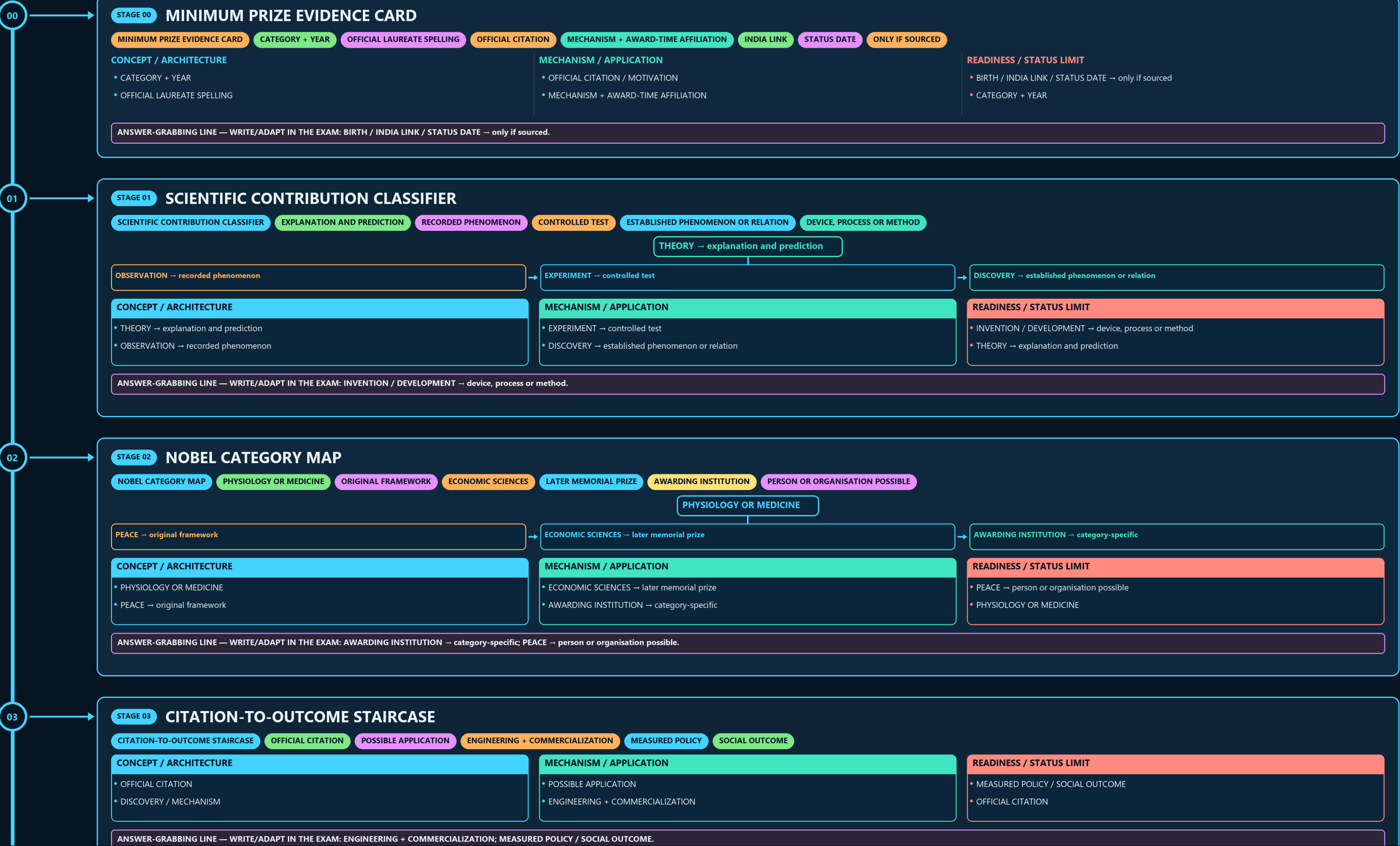
MINIMUM PRIZE EVIDENCE CARD → SCIENTIFIC CONTRIBUTION CLASSIFIER → NOBEL CATEGORY MAP → CITATION-TO-OUTCOME STAIRCASE → ATTRIBUTION AND CHRONOLOGY RAIL → PHYSICS 2025 QUANTUM-CIRCUIT MAP

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



PEACE → original framework

ECONOMIC SCIENCES → later memorial prize

AWARDING INSTITUTION → category-specific

CONCEPT / ARCHITECTURE

- PHYSIOLOGY OR MEDICINE
- PEACE → original framework

MECHANISM / APPLICATION

- ECONOMIC SCIENCES → later memorial prize
- AWARDING INSTITUTION → category-specific

READINESS / STATUS LIMIT

- PEACE → person or organisation possible
- PHYSIOLOGY OR MEDICINE

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: AWARDING INSTITUTION → category-specific; PEACE → person or organisation possible.

03

STAGE 03 CITATION-TO-OUTCOME STAIRCASE

CITATION-TO-OUTCOME STAIRCASE OFFICIAL CITATION POSSIBLE APPLICATION ENGINEERING + COMMERCIALIZATION MEASURED POLICY SOCIAL OUTCOME

CONCEPT / ARCHITECTURE

- OFFICIAL CITATION
- DISCOVERY / MECHANISM

MECHANISM / APPLICATION

- POSSIBLE APPLICATION
- ENGINEERING + COMMERCIALIZATION

READINESS / STATUS LIMIT

- MEASURED POLICY / SOCIAL OUTCOME
- OFFICIAL CITATION

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ENGINEERING + COMMERCIALIZATION; MEASURED POLICY / SOCIAL OUTCOME.

04

STAGE 04 ATTRIBUTION AND CHRONOLOGY RAIL

ATTRIBUTION AND CHRONOLOGY RAIL PRIOR THEORY LATER AND SEPARATE AWARD YEAR DISCOVERY YEAR AUTOMATICALLY

PRIOR THEORY → INSTRUMENT → EXPERIMENT

PUBLICATION → REPLICATION → AWARD

PATENT / PRODUCT / DEPLOYMENT → later and separate

LAUREATE != institution

AWARD YEAR != discovery year automatically

CONCEPT / ARCHITECTURE

- PRIOR THEORY → INSTRUMENT → EXPERIMENT
- PUBLICATION → REPLICATION → AWARD

MECHANISM / APPLICATION

- PATENT / PRODUCT / DEPLOYMENT → later and separate
- LAUREATE != institution

READINESS / STATUS LIMIT

- AWARD YEAR != discovery year automatically
- PRIOR THEORY → INSTRUMENT → EXPERIMENT

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PATENT / PRODUCT / DEPLOYMENT → later and separate.

05

STAGE 05 PHYSICS 2025 QUANTUM-CIRCUIT MAP

PHYSICS 2025 QUANTUM-CIRCUIT MAP SUPERCONDUCTING CIRCUIT + JOSEPHSON JUNCTION COLLECTIVE MACROSCOPIC STATE QUANTUM TUNNELLING SPECIFIC ENERGY AMOUNTS FAULT-TOLERANT QUANTUM COMPUTER

SUPERCONDUCTING CIRCUIT + JOSEPHSON JUNCTION

COLLECTIVE MACROSCOPIC STATE

BARRIER → quantum tunnelling

SPECIFIC ENERGY AMOUNTS → quantisation

CONCEPT / ARCHITECTURE

- SUPERCONDUCTING CIRCUIT + JOSEPHSON JUNCTION
- COLLECTIVE MACROSCOPIC STATE

MECHANISM / APPLICATION

- BARRIER → quantum tunnelling
- SPECIFIC ENERGY AMOUNTS → quantisation

READINESS / STATUS LIMIT

- DEMONSTRATION != fault-tolerant quantum computer
- SUPERCONDUCTING CIRCUIT + JOSEPHSON JUNCTION

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SPECIFIC ENERGY AMOUNTS → quantisation; DEMONSTRATION != fault-tolerant quantum computer.

06

STAGE 06 JOHN CLARKE CLUE RESOLVER

JOHN CLARKE CLUE RESOLVER CAMBRIDGE, UNITED KINGDOM UNIVERSITY OF CALIFORNIA, BERKELEY PHYSICS 2025 ONE-THIRD WITH DEVORET AND MARTINIS MATCH ALL CLUES THEN IDENTIFY

BORN → Cambridge, United Kingdom

AFFILIATION → University of California, Berkeley

CATEGORY / YEAR → Physics 2025

SHARE → one-third with Devoret and Martinis

CONCEPT / ARCHITECTURE

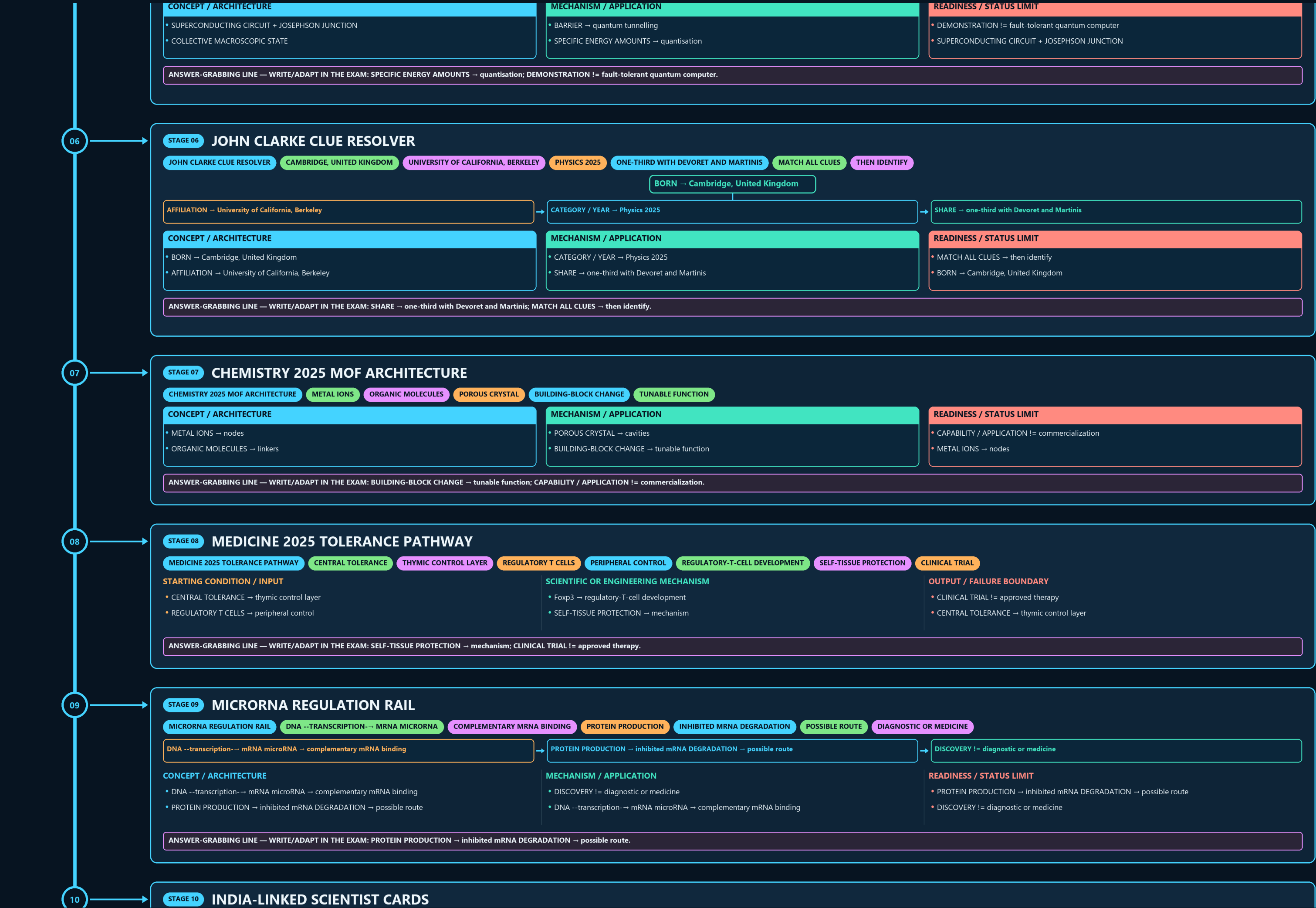
- BORN → Cambridge, United Kingdom
- AFFILIATION → University of California, Berkeley

MECHANISM / APPLICATION

- CATEGORY / YEAR → Physics 2025
- SHARE → one-third with Devoret and Martinis

READINESS / STATUS LIMIT

- MATCH ALL CLUES → then identify
- BORN → Cambridge, United Kingdom



ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SELF-TISSUE PROTECTION → mechanism; CLINICAL TRIAL != approved therapy.

09

STAGE 09 MICRORNA REGULATION RAIL

MICRORNA REGULATION RAIL

DNA --TRANSCRIPTION-- mRNA MICRORNA

COMPLEMENTARY MRNA BINDING

PROTEIN PRODUCTION

INHIBITED MRNA DEGRADATION

POSSIBLE ROUTE

DIAGNOSTIC OR MEDICINE

DNA --transcription-- mRNA microRNA → complementary mRNA binding

PROTEIN PRODUCTION → inhibited mRNA DEGRADATION → possible route

DISCOVERY != diagnostic or medicine

CONCEPT / ARCHITECTURE

- DNA --transcription-- mRNA microRNA → complementary mRNA binding
- PROTEIN PRODUCTION → inhibited mRNA DEGRADATION → possible route

MECHANISM / APPLICATION

- DISCOVERY != diagnostic or medicine
- DNA --transcription-- mRNA microRNA → complementary mRNA binding

READINESS / STATUS LIMIT

- PROTEIN PRODUCTION → inhibited mRNA DEGRADATION → possible route
- DISCOVERY != diagnostic or medicine

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PROTEIN PRODUCTION → inhibited mRNA DEGRADATION → possible route.

10

STAGE 10 INDIA-LINKED SCIENTIST CARDS

INDIA-LINKED SCIENTIST CARDS

RAMAN 1930

LIGHT SCATTERING

RAMAN EFFECT

KHORANA 1968

GENETIC CODE

PROTEIN SYNTHESIS

RAMAKRISHNAN 2009

CONCEPT / ARCHITECTURE

RAMAN 1930 → light scattering / Raman effect

KHORANA 1968 → genetic code / protein synthesis

MECHANISM / APPLICATION

RAMAKRISHNAN 2009 → ribosome structure and function

BIRTHPLACE != award-time affiliation

READINESS / STATUS LIMIT

CITATION != every later application

RAMAN 1930 → light scattering / Raman effect

CONCEPT / ARCHITECTURE

- RAMAN 1930 → light scattering / Raman effect
- KHORANA 1968 → genetic code / protein synthesis

MECHANISM / APPLICATION

- RAMAKRISHNAN 2009 → ribosome structure and function
- BIRTHPLACE != award-time affiliation

READINESS / STATUS LIMIT

- CITATION != every later application
- RAMAN 1930 → light scattering / Raman effect

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: KHORANA 1968 → genetic code / protein synthesis.

11

STAGE 11 PYQ AND FUTURE-STATUS FIREWALL

PYQ AND FUTURE-STATUS FIREWALL

2026 Q81

DIRECT BIOGRAPHICAL PRIZE ROUTE

BLUE LED

TOPIC 21 DIRECT SCIENCE OWNER

TOPIC 26

ATTRIBUTION + CITATION + MATURITY SUPPORT

2025 OFFICIAL AWARDS

2026 Q81 → direct biographical prize route

BOSE-EINSTEIN / BLUE LED → Topic 21 direct science owner

TOPIC 26 → attribution + citation + maturity support

2025 OFFICIAL AWARDS → latest completed science anchors

4 SEP 2026 → no invented 2026 laureates

SYSTEM / DEVICE / INSTITUTION

- 2026 Q81 → direct biographical prize route
- BOSE-EINSTEIN / BLUE LED → Topic 21 direct science owner

DECISIVE TECHNICAL DISTINCTION

- TOPIC 26 → attribution + citation + maturity support
- 2025 OFFICIAL AWARDS → latest completed science anchors

STATUS / UNIT / EVIDENCE LIMIT

- 4 SEP 2026 → no invented 2026 laureates
- 2026 Q81 → direct biographical prize route

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: 4 SEP 2026 → no invented 2026 laureates.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT

USE LAUREATE EXAMPLES TO ILLUSTRATE LONG-HORIZON PUBLIC RESEARCH, INSTRUMENTATION

MOBILITY AND THE PATH FROM BASIC SCIENCE TO STRATEGIC

AVOID "NOBEL COUNT" NATIONALISM.

A STRONGER ANSWER EXAMINES RESEARCH FUNDING, UNIVERSITIES, LABORATORY AUTONOMY

CONNECT TOPIC 24'S FUNDING ARCHITECTURE WITH THE SPECIALIST SCIENCE

SCIENCE & TECHNOLOGY

GS PAPER

OPTIONAL SCIENTIFIC / ENGINEERING DEPTH

- CAUTION: Use laureate examples to illustrate long-horizon public research, instrumentation, international
- mobility and the path from basic science to strategic technology.
- CAUTION: Avoid "Nobel count" nationalism.
- A stronger answer examines research funding, universities, laboratory autonomy, interdisciplinary teams and translation capacity.

READINESS, UNIT OR STATUS LIMIT

- CAUTION: Connect Topic 24's funding architecture with the specialist science file that explains the discovery.
- Subject: Science & Technology | Tier: Advanced | GS Paper: GS-III + Prelims.
- Purpose: Convert prize current affairs into deeper scientific reasoning without making optional enrichment necessary for objective-question coverage.
- Analytical lens CAUTION: Major prizes reveal how science moves from a counter-intuitive

COMPARATIVE TECHNOLOGY / GOVERNANCE NUANCE

- mechanism to an experimental demonstration and then to a technology platform.
- Deeper distinctions | Distinction | Why it matters | [---] | Discovery vs invention | A discovery establishes a phenomenon
- an invention creates a device or process.
- Prize citations may recognise either kind of contribution.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.